Name_____ Date_____ Period_____

**LESSON 2-7 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2**

# Dividing a Decimal by a Whole Number

Maryland Standard 6.NS.B.3 · Learning target – I can divide a decimal by a whole number and place the decimal point correctly.

**REMEMBER**

- Divide as you do with whole numbers. Put the decimal point in the quotient **straight above** the point in the dividend.
- If an amount is left over, **annex a zero** (write 0 after the last digit) and keep dividing.
- **Check:** quotient × divisor = dividend. Estimate first to catch a misplaced point.

**WORKED EXAMPLE**

Four friends share 22.2 kg of potting soil equally. How much does each get?

**Step 1** Estimate: 22 ÷ 4 is about 5.5. Put the point straight above.

**Step 2** 22 ÷ 4 = 5 R2. Then 22 tenths ÷ 4 = 5 tenths R2.

**Step 3** Annex a zero: 20 hundredths ÷ 4 = 5 hundredths.

**Answer** **5.55 kg each (5.55 × 4 = 22.2)**

**WORD BANK**

**dividend** – the number you share
**divisor** – the number you divide by
**quotient** – the answer to a division
**annex a zero** – write 0 at the end: 7.5 = 7.50

## PRACTICE

*show your work · label units*

**1.** Find each quotient.

**Step 1** Divide like whole numbers.

**Step 2** Put the point straight above the point in the dividend.

**a.** 9.6 ÷ 3 = _____ **b.** 18.5 ÷ 5 = _____ **c.** 2.94 ÷ 7 = _____

**d.** 0.56 ÷ 8 = _____ **e.** 31.2 ÷ 6 = _____ **f.** 9.3 ÷ 6 = _____

**2.** **Bulk Rice.** A **12-pound** bag of rice costs **$16.80**. Find the cost of **1 pound**.

**Step 1** Divide 16.80 by 12.

**Step 2** Check: multiply your answer by 12.

*Show your work.*

**Answer**_____ per pound

3. **Relay Race.** **8 runners** share a **10.2-kilometer** race equally. How far does **each runner** go?

**Step 1** Divide 10.2 by 8.

**Step 2** If there is a leftover, write 10.200 and keep dividing.

*Show your work.*

**Answer**_____ km

4. **Better Buy.** Store A sells 4 cans of beans for $3.56. Store B sells 6 cans for $5.10.

**a.** A, per can: $ ____ **b.** B, per can: $ ____ **c.** Better buy: ____

5. **Find the Error.** Marcus wrote 4.35 ÷ 5 = 8.7. Use an estimate to explain his mistake. Then give the correct quotient.

**Frame** *Marcus is wrong because 4.35 ÷ 5 is ___ than 1. The correct quotient is ___.*

| | | | |
|---|---|---|---|
| **CHECK YOURSELF** | ☐ I can do this on my own | ☐ I need a little help | ☐ I have a question about #___ |

**AT HOME** At the store, find a pack of several items. Divide the price by the number of items to find the cost of one.

Name_____ Date_____ Period_____

**LESSON 2-7 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4**

# Dividing by a Decimal

Maryland Standard 6.NS.B.3 · Learning target – I can divide by a decimal by making the divisor a whole number first.

**REMEMBER**

- Make the divisor whole: move **both** decimal points the **same** number of places right. The quotient does not change: 4.8 ÷ 0.6 = 48 ÷ 6.
- Annex zeros in the dividend if needed. Dividing by a number less than 1 gives a quotient **greater** than the dividend.

**WORKED EXAMPLE**

How many 0.15-liter cups can be filled from 4.5 liters of juice?

**Step 1** Make 0.15 whole: 0.15 × 100 = 15.

**Step 2** Do the same to 4.5: 4.5 × 100 = 450.

**Step 3** 450 ÷ 15 = 30. Check: 30 × 0.15 = 4.5

**Answer** **30 cups**

**WORD BANK**

**divisor** – the number you divide by

**power of ten** – 10, 100, 1,000

**equivalent** – has the same answer

## PRACTICE

*show your work · label units*

**1.** Rewrite with a whole-number divisor. Then divide.

**Step 1** Count the decimal places in the divisor.

**Step 2** Multiply both numbers by 10 or 100.

**Step 3** Divide the new problem.

**a.** 4.8 ÷ 0.6 = _____

**b.** 2.75 ÷ 0.5 = _____

**c.** 0.96 ÷ 0.12 = _____

**d.** 9 ÷ 0.3 = _____

**e.** 1.68 ÷ 0.04 = _____

**f.** 13.2 ÷ 1.1 = _____

**2.** **Pillows.** Leon has **7.5 yards** of fabric. **Each pillow** uses **1.25 yards**. How many pillows?

**Step 1** 1.25 has two decimal places, so multiply both numbers by 100.

**Step 2** Divide the new problem.

*Show your work.*

**Answer**_____ pillows

**3. Juice Glasses.** A pitcher holds 2.4 liters of juice. How many glasses can it fill?

**a.** 0.3-liter glasses: _____ **b.** 0.2-liter glasses: _____

**4. Walking Path.** One lap of a park path is 0.75 mile. How many laps make each distance?

**Step 1** Multiply both numbers by 100.

**Step 2** Divide.

**a.** 4.5 miles: _____ laps **b.** 6 miles: _____ laps

**5. Find the Error.** Keisha wrote 2.4 ÷ 0.08 = 24 ÷ 8 = 3. Explain her error. Find the correct quotient.

**Frame** *Keisha moved the divisor's point ___ places but the dividend's point ___ place. The correct quotient is ___.*

## TEST PRACTICE

*choose the best answer*

**6.** Which division has the same quotient as 6.72 ÷ 0.8?

**A** 672 ÷ 8 **B** 67.2 ÷ 8 **C** 6.72 ÷ 8 **D** 67.2 ÷ 80

**7.** How many 0.4-pound bags can be filled from 15.6 pounds of trail mix?

**A** 3.9 **B** 6.24 **C** 39 **D** 390

| CHECK YOURSELF | □ I can do this on my own | □ I need a little help | □ I have a question about #___ |
|---|---|---|---|

**AT HOME** Tell someone at home why 6 ÷ 0.5 is greater than 6.